

PAVAN ALAKUNTA

AI & Machine Learning Engineer

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PROFESSIONAL SUMMARY

Artificial Intelligence and Machine Learning graduate with hands-on expertise in designing, training, and deploying end-to-end machine learning and deep learning systems using Python, PyTorch, and TensorFlow. Proven ability to build production-grade computer vision and NLP pipelines. Strong mathematical foundations in linear algebra, probability, and optimisation. Seeking an AI/ML Internship or Engineer role to deliver scalable, high-impact AI solutions within a UK technology organisation.

TECHNICAL SKILLS

- **Programming Languages:** Python, C, R, MATLAB, SQL, JavaScript
- **ML Frameworks:** PyTorch, TensorFlow, Keras, Scikit-learn, Hugging Face Transformers
- **Computer Vision:** OpenCV, YOLOv8, YOLOv4
- **Natural Language Processing:** BERT, GPT fine-tuning, NLTK, Sentence Transformers, RAG Pipelines
- **Data Engineering:** Pandas, NumPy, Matplotlib, Seaborn, Plotly, PostgreSQL
- **Development Tools:** Linux (Ubuntu), Jupyter Notebook, VS Code, PyCharm
- **Mathematics:** Linear Algebra, Multivariate Calculus, Bayesian Inference, Probability Theory

WORK EXPERIENCE

LLM Training Team Lead | Centific, Hyderabad, India

2025

- Led and managed a team of 35 annotators and AI trainers to build, improve, and quality-assure training datasets for Large Language Models (LLMs) across NLP and generative AI applications.
- Managed end-to-end data annotation, labelling, and validation pipelines for NLP and generative AI tasks.
- Designed and implemented prompt engineering strategies to improve model response quality and above 98% accuracy.
- Conducted quality assurance (QA) and maintained high-quality datasets for LLM training.
- Executed RLHF (Reinforcement Learning from Human Feedback) workflows to fine-tune LLMs.
- Monitored KPIs including accuracy, consistency, and throughput of training data.

Freelance AI Trainer & Subject Matter Expert | Chegg / Outlier, Remote

2023 – 2024

- Provided expert-level solutions for complex technical and mathematical academic queries.
- Collaborated on AI training and data refinement projects for LLMs, ensuring 100% accuracy against strict deadlines.

AWS AI & ML Virtual Intern | AICTE – AWS Academy, India

2023

- Implemented cloud-based ML workflows covering data preprocessing, algorithm selection, and model deployment.
- Focused on building reproducible pipelines and foundational deployment practices on AWS.

KEY PROJECTS

Deepfake Video Detection using Deep Learning

2025

- Technologies: Python, TensorFlow, PyTorch, OpenCV, CNN/LSTM, FaceForensics++ dataset.
- Implemented a hybrid CNN-LSTM architecture to detect spatial and temporal inconsistencies in deepfake videos.
- Trained on FaceForensics++ and Celeb-DF datasets using transfer learning with an EfficientNet backbone.
- Deployed as a Flask web application with real-time video processing capability.

Crop Protection from Animal Intrusion using Machine Learning

2024

- Technologies: Python, TensorFlow/Keras, OpenCV, Raspberry Pi, IoT Sensors.
- Developed a real-time animal intrusion detection system using CNNs to prevent crop damage.
- Trained a Convolutional Neural Network on custom datasets of wild boar, deer, and other animal imagery.
- Integrated ML model with IoT sensors and Raspberry Pi for edge deployment.

EDUCATION

MSc Artificial Intelligence and Robotics | University of Hertfordshire, Hatfield, UK *Jan 2026 – Present*

- Key Modules: Artificial Life with Robotics, Neural Networks and Machine Learning, Theory and Practice of Artificial Intelligence.
- Activities: University AI and Robotics Society.

B.Tech Artificial Intelligence and Machine Learning | Nalla Malla Reddy Engineering College, Hyderabad, India *Oct 2021 – May 2025*

- Classification: First Class Honours | GPA: 7.68 / 10.00
- Final Year Project: Deepfake Video Detection using Deep Learning.
- Core Modules: Data Structures and Algorithms, OOP, Linear Algebra, Probability and Statistics, Computer Networks, AI, Machine Learning, Deep Learning, NLP, Software Engineering.

CERTIFICATIONS

- AWS AI/ML Virtual Internship – AICTE | Completed 2023

ADDITIONAL INFORMATION

- **Languages:** English (Fluent), Telugu (Native), Hindi (Proficient)
- **Right to Work:** UK Student Visa holder – eligible for UK Graduate Route Visa providing 2 years post-study work authorisation upon graduation.
- **Interests:** Large Language Models and Generative AI, AI Ethics and Model Evaluation, Emerging AI Technologies, Cooking, Basketball.